

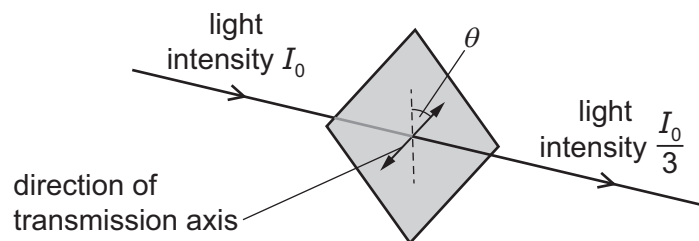
- 26 A buzzer emitting sound of frequency 846 Hz is attached to a string and rotated in a horizontal circle. The linear speed of the buzzer is 25.0 m s^{-1} .



The speed of sound is 340 m s^{-1} .

What is the maximum frequency heard by the observer?

- A** 783 Hz **B** 788 Hz **C** 908 Hz **D** 913 Hz
- 27 Vertically polarised light of intensity I_0 is incident normally on a polarising filter.
- The transmission axis of the filter is at an angle θ to the plane of polarisation of the light.
- The intensity of the light after passing through the filter is $\frac{I_0}{3}$.



What is θ ?

- A** 8.4° **B** 55° **C** 71° **D** 84°
- 28 Which statement compares the behaviour of transverse and longitudinal waves?
- A** Only longitudinal waves can be diffracted.
- B** Only longitudinal waves can travel in free space.
- C** Only transverse waves can be coherent.
- D** Only transverse waves can be polarised.